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ORIGINAL ARTICLE

EEG alpha and theta time-frequency structure during a written mathematical task

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Abstract
Since the first electroencephalogram (EEG) was obtained, there have been many possibilities to use it as a tool to access brain cognitive dynamics. Mathematical (Math) problem solving is one of the most important cortical processes, but it is still far from being well understood. EEG is an inexpensive and simple indirect measure of brain operation, but only recently has low-cost equipment (mobile EEG) allowed sophisticated analyses in non-clinical settings. The main purpose of this work is to study EEG activation during a Math task in a realistic environment, using mobile EEG. A matching pursuit (MP)-based signal analysis technique was employed, since MP properties render it a priori suitable to study induced EEG activity over long time sequences, when it is not tightly locked to a given stimulus. The study sample comprised sixty healthy volunteers. Unlike the majority of previous studies, subjects were studied in a sitting position with their eyes open. They completed a written Math task outside the EEG lab, wearing a mobile EEG device (EPOC+). Theta [4 Hz–7.5 Hz], alpha (7.5 Hz–13 Hz] and 0.5 Hz micro-bands in the [0.5 Hz–20 Hz] range were studied with a low-density stochastic MP dictionary. Over 1-min windows, ongoing EEG alpha and theta activity was decomposed into numerous MP atoms with median duration around 3 s, similar to the duration of induced, time-locked activity obtained with event-related (des)synchronization (ERS/ERD) studies. Relative to Rest, there was lower right-side and posterior MP alpha atom/min during Math, whereas MP theta atom/min was significantly higher on anteriorly located electrodes, especially on the left side. MP alpha findings were particularly significant on a narrow range around 10 Hz–10.5 Hz, consistent with FFT alpha peak findings from ERS/ERD studies. With a streamlined protocol, these results replicate previous findings of EEG alpha and theta activation obtained during Math tasks with different signal analysis techniques and in different time frames. The efficient application to real-world, noisy EEG data with a low-resolution stochastic MP dictionary shows that this technique is very encouraging. These results provide support for studies of mathematical cognition with mobile EEG and matching pursuit.

Keywords Emotive EPOC · EEG microstructure · Theta activity · Matching pursuit · Math cognition · Mobile EEG

1 Introduction

Although it is not new that the central nervous system (CNS) evaluates quantities in a variety of ways, it is in accuracy that we humans perceive good evolutionary advantage [1–7]. It is therefore not surprising that numerical cognition has become a matter of deep interest when one considers how brain cells produce thought or action. As the gap between neurophysiology and cognitive neuroscience is bridged through growing knowledge of detailed CNS circuitry and function, it has become apparent that real-world tasks demanding number manipulations will recruit a very specific subset of neurons bilaterally located in the intraparietal sulcus, but they will not be satisfactorily carried out without involving multiple brain sites such as the left posterior temporal language area and bilateral prefrontal and inferior temporal regions, and multiple cognitive abilities such as attention, working memory and visuo-spatial skills [8, 9]. These findings imply a degree of functional vs. topological multiplicity that needs to be considered in theories regarding cognitive strategies applied to arithmetic problem solving [3]. Procedural and memory retrieval strategies are believed to be employed in alternate and complementary ways under the influence of individual preference, cognitive structure, task difficulty, degree of skill,

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task commitment and developmental stage [10, 11]. Reliance on retrieval as opposed to procedural strategies is thought to increase as one becomes more familiar with operatory functions and arithmetic manipulations [12]. Systematic testing of these theories, however, is still a work in progress [13, 14]. Contemporary students, at least in Western countries, have possibly not been exposed to the amount of mathematical fact learning by repetition to which older generations have been submitted [15]. It is not clear whether or to what extent differences in learning paradigms will impact how the mature brain deals with arithmetic. As Mathematics (Math) teaching policies across different educational programs are the subject of heated debate and learning experiences continue to extrapolate the context of traditional classrooms, practical methods to assess Math cognition in realistic settings will expectedly gain importance in ensuing years [15–18].

One source of validation for the concept of different brain strategies involved in Math cognition comes from electrophysiology studies that show how electroencephalography (EEG) rhythms may be elicited over different scalp locations employing a variety of cognitive manipulations [19, 20]. Predominantly left-sided EEG theta activation (synchronization, power increase) has been linked with Math fact retrieval, while widespread EEG alpha activation (desynchronization, power decrease) has been associated with procedural numerical manipulations, perhaps through more intense recruitment of working memory and concentration [19, 20]. EEG frequency analysis is non-invasive, relatively inexpensive, easy to conduct in large subject groups and made more attractive by the introduction of mobile EEG devices, currently undergoing technical troubleshooting and testing in naturalistic environments such as classrooms and outdoors [21–24]. Extensive information on EEG spectral content has been obtained with standard and high-density EEG equipment over the past decades. However, variability in experimental paradigms has critically limited inter-study comparisons across time. Seminal contributions stem from a heroic era when digital signal acquisition was nonexistent and statistical analysis was incipient [6, 25–30]. Many conclusions have been reached through indirect theoretical constructs, for instance, EEG frequency ratios obtained for cognitive tasks considered at the time to specifically engage opposed brain hemispheres, or computed for groups contrasted by performance level, prone to influence by many factors, or ascribed to categories of cognitive style based on psychological self-evaluation [19, 20, 31–33]. Of direct importance to the present study, previous information has been mainly obtained with Math tasks carried out with eyes closed, sometimes with subjects lying down on the supine position, not necessarily reproducible when subjects are on a sitting position with eyes open.

The ongoing EEG is highly non-stationary [34]. Frequency analysis carried out over different time frames may yield divergent results, and different signal analysis techniques may better suit investigation on different time scales. With classical and most widely used spectral analysis by Fourier transform, changes in EEG power spectra can be reliably identified in association with specific stimuli in the close range of 0.5 s, as shown for event-related desynchronization (ERD) and synchronization (ERS) studies [35, 36]. This degree of time resolution is achieved at the expense of frequency resolution [37]. During sustained cognitive activity, short-lasting changes will no longer be discernible, even though long-lasting, consistent changes on narrow frequency bands may be identified. Under this rationale, it has been concluded that induced cognitive EEG activity, on the timescale of seconds, will more closely resemble a resting (reference) period than shorter-lasting, event-related EEG epochs [38, 39]. However, when attempting to characterize ongoing EEG microstructure on the range of seconds, it may be interesting to explore a different strategy such as the matching pursuit algorithm, which makes use of a large dictionary of different mathematical functions to attain a description of the original signal [37]. Matching pursuits are greedy algorithms, meaning that the original signal is decomposed through recursive steps called iterations whereby the single dictionary function that best matches the signal is identified and extracted, leaving a progressively less meaningful signal residue to undergo the next iteration. It resembles the strategy employed to find the smallest possible number of coins when giving change in a money transaction. The result is a set of functions that can be represented in a time-frequency plane as elements called atoms, defined through their frequency, length and intensity [37] (see Fig. 1). This is akin to describing a melody through its written score, where musical notes are denoted on a time-frequency plane, with frequency (pitch) indicated on the y axis. Unlike the melody, however, a "reverse procedure" would only recover main characteristics instead of the entire original signal, since matching pursuit is an adaptive approximation technique. Practical applications include digit and face identification and image reconstruction [40–42]. Among other algorithm variations, an adaptation for large-size data like EEG (consisting in the replacement of the original dyadic dictionary by a stochastic dictionary of functions to avoid statistical bias in the decomposition) has been provided by [43]. Matching pursuit with stochastic dictionary of functions has been successfully applied to the description of sleep EEG microstructure and wake EEG alpha activity [43–49]. To our knowledge, EEG changes induced by arithmetic calculations have not been studied with matching pursuit procedures.

In this study, EEG alpha and theta microstructure during a sustained arithmetic production task was investigated with a streamlined experimental protocol. Subjects wearing a mobile EEG device were tested outside the EEG lab, performing a written task, in the sitting position with eyes open. A matching pursuit algorithm was employed for signal anal-

ysis, since its properties render it a priori suitable to study induced EEG activity over long time sequences, when it is not tightly locked to a given stimulus. The main goal was to verify if previous results obtained in tighter experimental protocols, over shorter time frames and employing different signal analysis techniques could be replicated through this approach. Since the Math test consisted of a multiplication task thought to require several cognitive abilities, no theoretical assumptions regarding strategy or specific cognitive functions were implied [4, 50].

2 Methods

2.1 Subjects and experimental protocol

Sixty healthy volunteers (mean age 21.9, standard deviation 3.4, 29 males) recruited among local undergraduate students and research staff with similar age distribution [51] took part in the study. Manual preference and menstrual cycle were not controlled. The protocol complied with good clinical practice and the Helsinki declaration [52]. Written informed consent was provided by all participants. The study was approved by the local ethics committee (Faculdade de Medicina de Botucatu - UNESP) under project number 59229916.4.0000.5411.

Subjects were comfortably seated in a non-shielded room with natural lighting. Individual 20-min sessions alternated 10 times between two 1-min conditions: eyes open (Rest) and Math task (Math). Subjects were instructed not to talk and to remain as still as possible. Blinking was allowed. During Math, a four-by-four-digit multiplication test was to be solved on a blank sheet of paper. Math problems were all different, randomly generated and of similar complexity. Feedback regarding performance was provided after total completion. Score for each Math problem ranged from 0 to 10 as each correctly solved line (4 multiplications plus 1 sum) was worth 2 points. A final average score from 0 to 10 was ascribed to each subject. There was overall good performance with median score 7, mode 6. Figure 1 illustrates and summarizes the protocol.

2.2 EEG device and data acquisition

EEG was acquired on a 16-bit A/D converter, Emotiv EPOC+ portable device [21], that uses gold-plated contact sensors fixed to flexible plastic arms of a wireless headset (http://emotiv.com/). EEG is acquired from left and right anterior frontal (AF3, AF4), anterior temporal (F7, F8), frontal (F3, F4), fronto-central (FC5, FC6), temporal (T7, T8), parietal (P7, P8), occipital (O1, O2) and mastoid (M1, M2) electrodes aligned with 10-10 IS (International System) [34], exception for T7 and T8, which are placed more posteriorly on 10-10 IS. As it is a fixed headset, there are variations on electrode position according to head size and shape which were not controlled. M1 functioned as a reference point to which amplitude of every other sensor was compared. M2 was a feed-forward reference that reduced external electrical interference. Room lights and, when necessary, other electrical devices were turned off to avoid 60 Hz noise interference. Signal acquisition was only carried out when EPOC+ default impedance check showed acceptable (level 4) electrode impedance.

2.3 Spectral power and MP signal decomposition

Signals were internally digitized at 2048 Hz, down-sampled to 128 Hz and filtered between 0.16 Hz and 43 Hz at the acquisition module before transmission [21]. The headset was wirelessly connected to a PC and interfacing SDK Emotiv platform. Signals were then converted to ASCII. Spectral analysis was carried out with TISEAN 3.0.1 [53] default Power spectrum feature, which yields square-windowed, standard periodograms, implemented within a customer-built pipeline. Sampling rate was 128 Hz. Each 1-min EEG segment (7680 digital points) was zero-padded to achieve 64 s (8192 digital points, a power of two). Frequency resolution was 0.016 Hz. Spectral analysis was carried out to provide an overview of main signal characteristics. It was not directly compared to MP. Data were visually checked for the presence of noise. Periodograms containing power spectral density (PSD) peaks exceeding an arbitrary threshold of 1000 μV^2/Hz were discarded. These comprised less than 3% of the total sample. Band-limited theta and alpha power were respectively defined as average absolute band power in [4 Hz–7.5 Hz] and (7.5 Hz–13 Hz] bands, expressed in μV^2. As band limits are arbitrary, values were also computed for each 0.5 Hz interval within the 0.5 Hz–20 Hz range.

The matching pursuit algorithm employed in this study, hereafter referred to as MP, is freely available at https://github.com/dellagustin). It is a customized version of another freely available algorithm (called MP, version mp3) which was introduced by Durka and cols. in [43] and remained available at https://braintech.pl/eeg-pl/ until 2015. The algorithm MP, version mp3 differs from the original algorithm developed by Mallat and Zhang [37] in that a stochastic dictionary of functions replaced the original dyadic dictionary, to avoid statistical bias in the decomposition. The stochastic dictionary in MP, version mp3 is basically built from real Gabor functions, with discrete Dirac and Fourier functions added to deal with well-localized structures in time and frequency, respectively [43]. This dictionary choice is not unique as explained in [54]. Our customization of MP, version mp3 consisted in the further inclusion of a Gabor chirplet dictionary of functions as a subroutine which allows the assessment

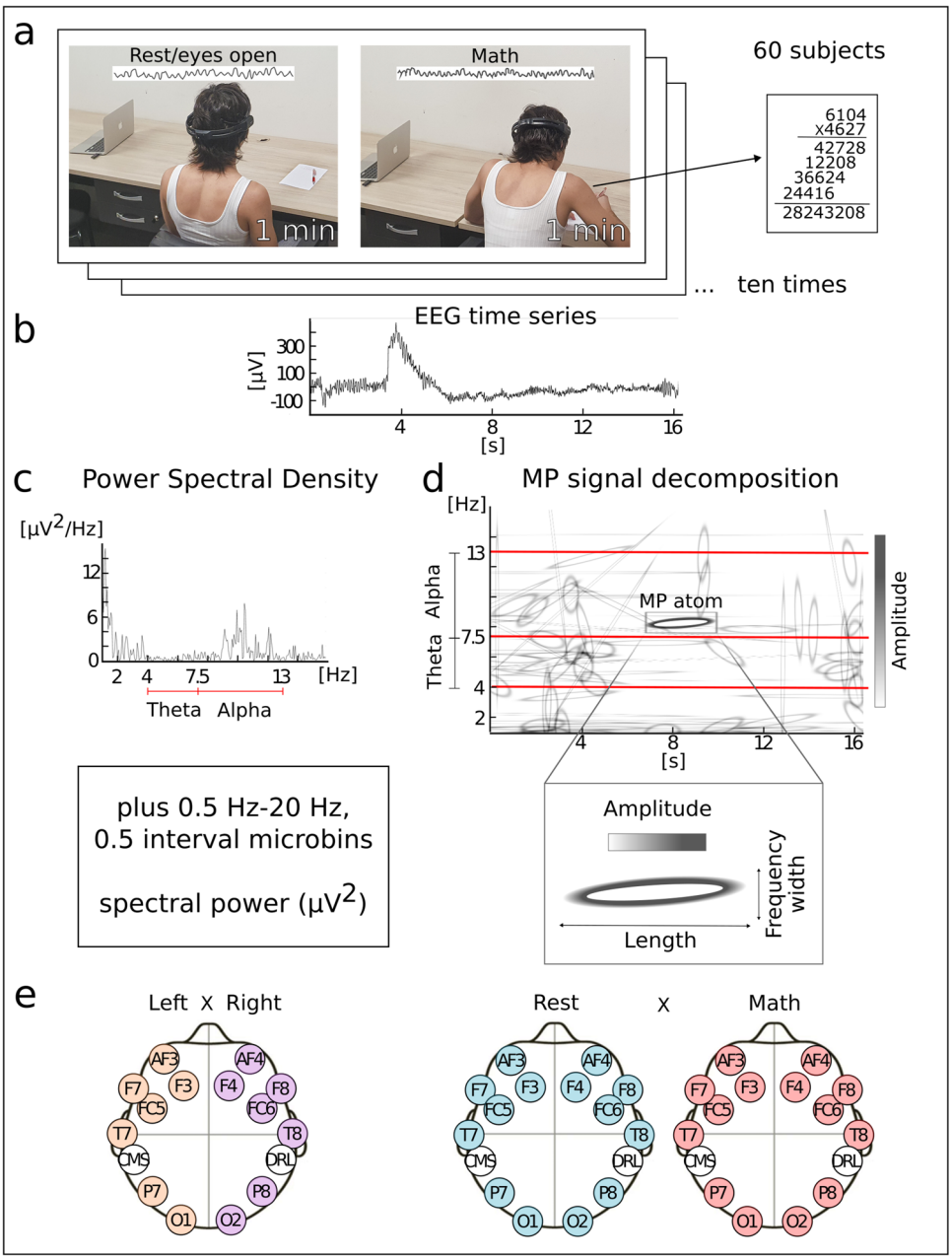

Fig. 1 Overview of study protocol. a Data acquisition procedure with mobile EEG. Rest with eyes open (1 min) followed by four-by-four-digit multiplication test (1 min). Steps repeated ten times for 60 subjects. b Illustration of an actual 16 s-EEG segment. Eyelid movement deflection near the fourth second. Alpha activity visually discernible on initial and final portions of the time series. c Illustration of power spectral density distribution for the same 16 s time series depicted in b. Frequency represented in x axis. Energy represented in y axis. Power peaks noticeable on the alpha range. (Notice that actual analyses were carried out in 1-min EEG windows, see text for details.) Average absolute band-limited spectral power computed in theta [4 Hz–7.5 Hz] and alpha (7.5 Hz–13 Hz] bands, plus 0.5 Hz microbins in the 0.5 Hz–20 Hz range. d Matching pursuit (MP) signal decomposition for the same EEG segment depicted in b, represented by atoms in the time-frequency plane (Wigner distribution). Time is in x axis, frequency is in y axis and MP atom amplitude (intensity) is illustrated in gray color scale, with atoms denoted as "hollow ellipses" to avoid visual pollution. e Statistical comparisons carried out for hemisphere side (left vs. right) and group condition (Rest vs. Math)

of atom internal frequency modulation [55, 56]. We describe this implementation in more detail in [46].

MP projects the signal $Sg(t)$ on a vector g_γ

$$g_\gamma(t) = \alpha e^{-\pi((t-b)/s)^2} \sin[2\pi\omega(t-b)/N + (2\pi)^2\xi(t-b)^2/N^2 + \phi], \quad (1)$$

where t is time, N is the signal digital point number (signal size), b is position in time, s is scale, ω is angular frequency, ξ is the chirping term and ϕ is phase. $\gamma = \{\alpha, b, s, w, \xi, \phi\}$ is a parameter list for dictionary functions where α is selected such that $||g_\gamma|| = 1$.

MP was implemented into successive juxtaposed windows over the time series, run separately for each EEG channel. Each window had 2048 points and 16 s duration. The fourth window from each minute was 12 s long, so it was zero-padded to achieve 2048 points. The original dictionary size was 70.000 (this number is inflated at the subroutine level by the presence of chirplet functions). A stochastic dictionary size of 70 K/16 s may be considered small, as early MP implementations for EEG signal analysis relied on stochastic dictionary sizes in the range of millions [57]. However, the low-density dictionary used here has been empirically shown to perform well to detect intermittent, short-duration sigma activity (sleep spindles, ≤2 s, 11 Hz–16 Hz) embedded on highly non-stationary background rhythms, and to discriminate between different severity sleep apnea groups according to sleep spindle characteristics [45, 47, 58]. In intracranially acquired signals, a dictionary size of 100 K/2 s was sufficient to test MP performance for epileptic seizure state discrimination, where frequency is known to vary widely within a short period of time [59]. As we were dealing with highly filtered, intermediate/low-frequency scalp EEG activity acquired on a low-resolution mobile device, we expected distinctive characteristics of such a signal to be retrieved over the first iterations with a low-density dictionary. The stopping criterion was established at 96 iterations. Since the decomposition easily attains 90% signal energy, 96 iterations can only be reached by setting the energy parameter threshold to 100%. MP decomposition thus obtained 96 atoms for each 16 s window in each of 14 EEG channels. This methodology tends to enforce constant global density, so it differs from measurements of atom density such as described in [60].

Atoms obtained with MP have a central point in the time-frequency plane and limits determined by their half-width (HW), corresponding to $\pm\sigma$ on a Gaussian curve [37]. Each atom is identified through its central point in time. Atom frequency corresponds to its central point in frequency. Atom length corresponds to HW in time. Atom peak amplitude corresponds to the AT parameter in the algorithm. Data pertaining to atom count, single atom amplitude (μV), length (s) and central frequency (Hz) were collected. Internal frequency modulation was computed but not analyzed here.

2.4 Statistical analysis

After MP decomposition, theta [4 Hz–7.5 Hz] and alpha (7.5 Hz–13 Hz] atoms with ≥ 0.5 s duration and ≤ 250 μV amplitude were selected for further analysis. Single alpha and theta distributions were calculated for each 10-min dataset obtained from each of 60 subjects, in 2 conditions (Rest and Math), for each of 14 electrodes. From these single distributions, the 10% lower amplitude alpha and theta atoms were respectively discarded as additional noise control.

Variables under investigation were MP atom number/min (count), duration in s (length), central frequency in Hz (Freq), amplitude in μV (amplitude), and theta and alpha spectral power in μV^2 (power). Because frequency bands have arbitrary limits and often differ among studies, MP atom count and amplitude, as well as spectral power, were also studied in 0.5 Hz micro bins within the [0.5 Hz–20 Hz] range. MP atom length and frequency could not be analyzed in 0.5 Hz bands due to intrinsic properties of MP description. When frequency slices start to decrease, time and frequency become more diffusely defined. As time is known, frequency is unknown [37, 61].

Shapiro-Wilk tests for normalcy revealed non-Gaussian distributions, so analyses proceeded with non-parametric methods. Our null hypothesis is that EEG alpha and theta time-frequency structure, as measured through a matching pursuit procedure with the present experimental protocol, is similar in rest and mental calculation states. Comparisons between left and right hemispheres and between Rest and Math were implemented with two-tailed Wilcoxon rank-sum tests (W). False discovery rate (FDR) -corrected P-values (q-values) < 0.05 were considered significant. Unless otherwise specified, data are expressed as median [interquartile range, IQR] and depicted as notched Box-and-Whisker plots [62], where notches represent 95% confidence intervals for medians and whiskers represent standard extreme values, respectively, 1.5 times above the upper quartile and 1.5 times below the lower quartile. Statistical analyses were run using R version 3.6.2 [63, 64].

3 Results

3.1 Alpha and theta time-frequency structure in Rest EEG

Table 1 summarizes values obtained with the acquisition parameters set in this study. These data may be visually inspected for the presence of amplitude and frequency topography gradients, well-known to the routine EEG reader [34].

Table 1 EEG spectral power and MP atom descriptors in the alpha and theta range during Rest and Math

Freq Range	Side	Region	Rest	Power [$\times 10^{-2}\mu v^2$]	AT [μV]	Atoms [♯/min]	Freq [Hz]	Length [s]	Math	Power [$\times 10^{-2}\mu v^2$]	AT [μV]	Atoms [♯/min]	Freq [Hz]	Length [s]
Theta	Left	Anterior frontal	AF3	6.9	41.76	40.6	5.47	2.64	AF3	11	45.74	47.52	5.47	2.77
	Right	Anterior frontal	AF4	11	**46.72**	43.35	5.44	2.69	AF4	17	**53.14**	48.4	5.5	2.83
	Left	Frontal	F3	4.1	33.89	43.05	5.5	2.88	F3	5	36.32	47.53	5.5	2.91
	Right	Frontal	F4	5.2	39.13	43.75	5.5	2.88	F4	5.6	39.28	47.15	5.5	2.94
	Left	Anterior temporal	F7	7.1	41.46	39.6	5.38	2.66	F7	7.7	43.44	42.88	5.44	2.84
	Right	Anterior temporal	F8	12	**52.07**	**42.29**	5.44	2.71	F8	12	**53.75**	43.65	5.44	2.91
	Left	Fronto-central	FC5	4	32.9	34.4	5.44	2.62	FC5	4.3	32.35	40.02	5.44	2.86
	Right	Fronto-central	FC6	**7.7**	**42.7**	**38.58**	5.44	2.80	FC6	**6.3**	**43.53**	39.9	5.44	2.89
	Left	Temporal	T7	3	30.54	31.96	5.44	2.84	T7	3.7	31.74	35.2	5.38	2.84
	Right	Temporal	T8	**6.5**	**47.36**	**37.1**	5.44	2.91	T8	**6.3**	**51.04**	32.25	5.38	2.94
	Left	Parietal	P7	2.1	25.57	35.95	5.44	2.83	P7	2.8	28.76	36.5	5.44	2.83
	Right	Parietal	P8	**5**	**40.11**	37.67	5.44	2.77	P8	**5.1**	**43.11**	34.7	5.44	2.93
	Left	Occipital	O1	2.6	27.15	37.84	5.5	2.84	O1	2.4	28.61	39.95	5.44	2.98
	Right	Occipital	O2	**3.8**	**36.6**	37.62	5.44	2.78	O2	**4.9**	**38.29**	38.95	5.44	2.88
Alpha	Left	Anterior frontal	AF3	3.9	38.23	42.7	9.91	2.90	AF3	3.1	37.71	40.8	9.57	3.00
	Right	Anterior frontal	AF4	6.5	**45.15**	41.2	9.94	2.88	AF4	**5.2**	**44.27**	40.29	9.66	3.00
	Left	Frontal	F3	2.8	31.18	45.55	9.97	3.02	F3	2	30.71	41.55	9.69	3.10
	Right	Frontal	F4	3.2	34.92	48.65	9.94	3.02	F4	2.3	32.92	43.2	9.75	3.20
	Left	Anterior temporal	F7	4.3	38.29	37.8	9.88	2.93	F7	2.8	36	35.1	9.69	2.88
	Right	Anterior temporal	F8	5.8	**48.09**	41.39	10.01	3.06	F8	**4.3**	**46.04**	**39.65**	9.79	3.00
	Left	Fronto-central	FC5	2.8	30.23	35.3	9.91	2.93	FC5	1.9	27.57	33.47	9.82	3.14
	Right	Fronto-central	FC6	**4.8**	**42.33**	**43.2**	10.1	2.97	FC6	**3.2**	**38.55**	**40**	9.88	3.05
	Left	Temporal	T7	2.2	28.39	34.08	9.97	2.90	T7	1.7	27.59	31.8	9.88	3.06
	Right	Temporal	T8	**5.1**	**44.79**	**44.3**	10.26	3.00	T8	**4.4**	**47.37**	**38.05**	10.13	3.09
	Left	Parietal	P7	1.5	24.88	40.3	10.04	3.03	P7	1.5	25.45	34.45	9.91	3.07
	Right	Parietal	P8	**4.8**	**41.14**	**44.45**	10.26	3.02	P8	**4.5**	**40.11**	36.94	10.01	3.00
	Left	Occipital	O1	1.8	25.97	46.35	10.13	2.97	O1	1.3	26.66	40.6	9.94	3.12
	Right	Occipital	O2	**4.2**	**36.8**	49.6	10.26	2.95	O2	**2.5**	**33.08**	40.75	10.01	3.16

Statistical differences (two-tailed Wilcoxon rank-sum test, FDR-corrected p-values < 0.05) between hemispheres, higher values (always on the right side) denoted in boldface. AT: MP atom amplitude

In Rest, alpha and theta power shows a clear trend toward higher amplitude in more anterior as opposed to more posterior regions. The anterior-posterior amplitude gradient was also evident for MP amplitude. Equally expected [34, 65], absolute power and MP amplitude were generally higher on the right side compared to the left side (Appendix A Fig. 5). Side comparisons for MP amplitude reached significance for all locations under study, except frontal (F3 vs F4) sites, for alpha as well as theta activity. When measured through spectral power, left vs. right side amplitude differences were significant in mid and posterior locations both for alpha and for theta activity.

Considerably different patterns were seen for MP atom number distribution according to brain topography and frequency range. There were no clear trends or antero-posterior gradient on the left hemisphere. The right hemisphere still showed a trend toward higher median theta count on anterior as opposed to posterior sites. On the alpha count, the right hemisphere showed a trend toward an inverse (posterior-anterior) gradient. Inter-hemisphere atom number comparisons showed similar patterns to power and MP amplitude, i.e., a trend toward higher numbers on the right side. However, significantly higher right-sided counts were only found for theta on fronto-central and midtemporal regions, and for alpha on the midtemporal region.

MP frequency and length were notably stable across brain topography during Rest. Theta length ranged between 2.62 [0.62] s on FC5 and 2.91 [0.44] s on T8, whereas alpha length ranged between 2.88 [0.63] s on AF4 and 3.06 [0.48] s on F8, respectively. No anterior-posterior gradients or inter-hemisphere differences were identified, except for higher alpha frequency on the right midtemporal region compared to the left side.

3.2 Amplitude distribution during Math vs. Rest

There were no statistical differences in alpha and theta amplitude between Rest and Math conditions, either through spectral power or through MP atom amplitude comparisons.

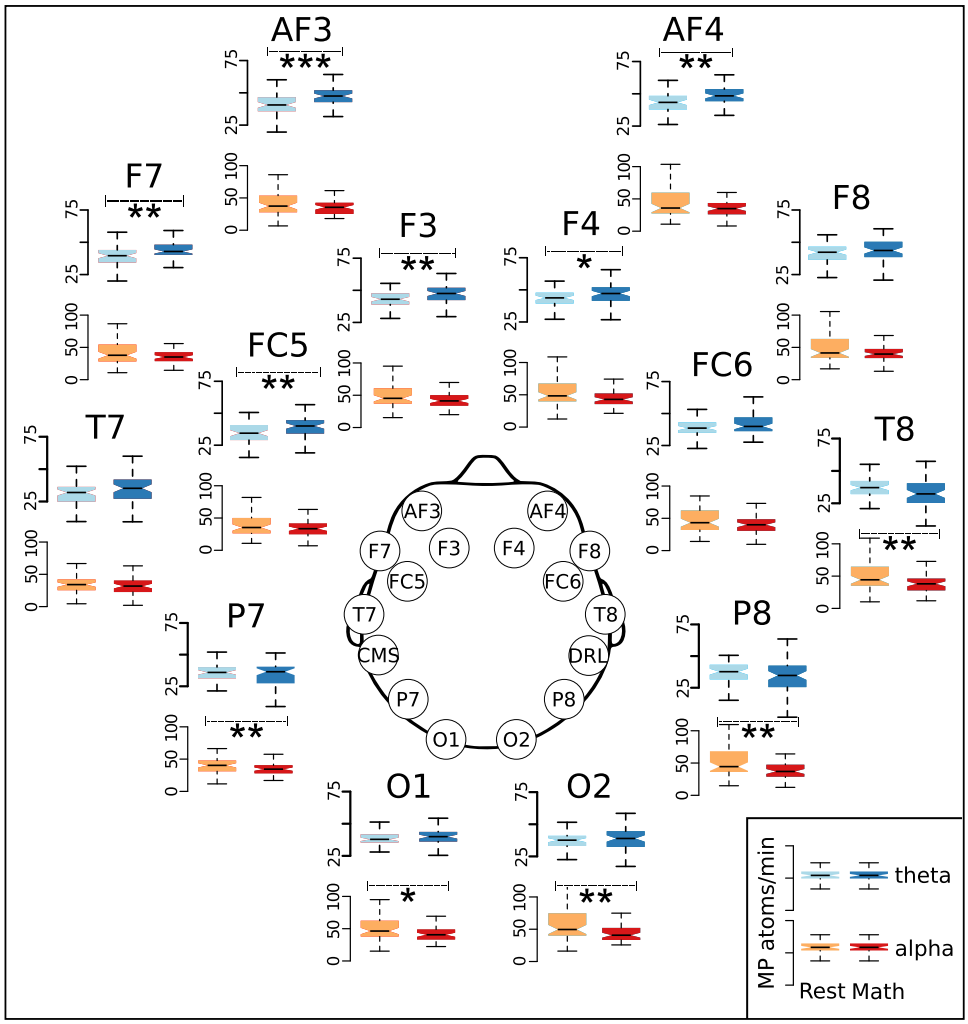

Fig. 2 MP alpha and theta atoms/min according to brain topography (scalp EEG electrodes) during Rest vs. Math. Electrode positions depicted at the figure center for better visualization. Odd numbers corresponding to left hemisphere electrodes. Two-tailed Wilcoxon rank-sum tests. FDR-corrected p-values (q-values) $* \, p \leq .05$; $** \, p \leq .01$; $*** \, p \leq .001$

3.3 Number of alpha and theta MP atoms during Math vs. Rest

Figure 2 shows alpha and theta MP atom numbers during Math vs. Rest. Compared to Rest, Math theta count was significantly higher on all electrodes pertaining to the left anterior head section. The same trend was found for the right anterior head section, where differences reached statistical significance on anterior frontal and frontal sites. On posterior (parietal and occipital) regions, there were no such findings on either hemisphere for the theta range.

Whereas theta count differences were noticeable on anterior brain sites and apparently stronger on the left side, alpha count differences were more conspicuous on posterior and right-side regions, despite being better balanced between hemispheres. Moreover, alpha count findings, when contrasted to theta count, went in the opposite direction. Compared to Rest, Math alpha count was significantly lower on right posterior temporal and bilateral parieto-occipital regions.

Inter-hemisphere atom number comparisons on the theta range were all non-significant. Alpha count still showed a trend toward higher values on the right side compared to contralateral areas, which only reached statistical significance on fronto-central regions.

3.4 MP atom frequency during Math vs. Rest

Figure 3 shows MP alpha frequency distribution during Math and Rest. Compared to Rest, Math alpha atoms had lower median frequency in most locations. Statistical significance was reached for every electrode on the anterior region, on left parietal and on left and right occipital regions.

Math alpha frequency showed a trend toward lower values on left anterior, as opposed to right posterior regions, so that it ranged from 9.57 [0.50] Hz on AF3 to 10.13 [0.58] Hz on T8. There were no significant inter-hemisphere differences except for higher frequency on the right temporal region.

On the theta range, frequency distribution during Math was not statistically different from Rest. There were no inter-hemisphere differences and no clear antero-posterior gradients.

3.5 MP atom duration during Math vs. Rest

MP atom length comparisons between Rest and Math conditions are depicted in Fig. 6, Appendix B. A statistically

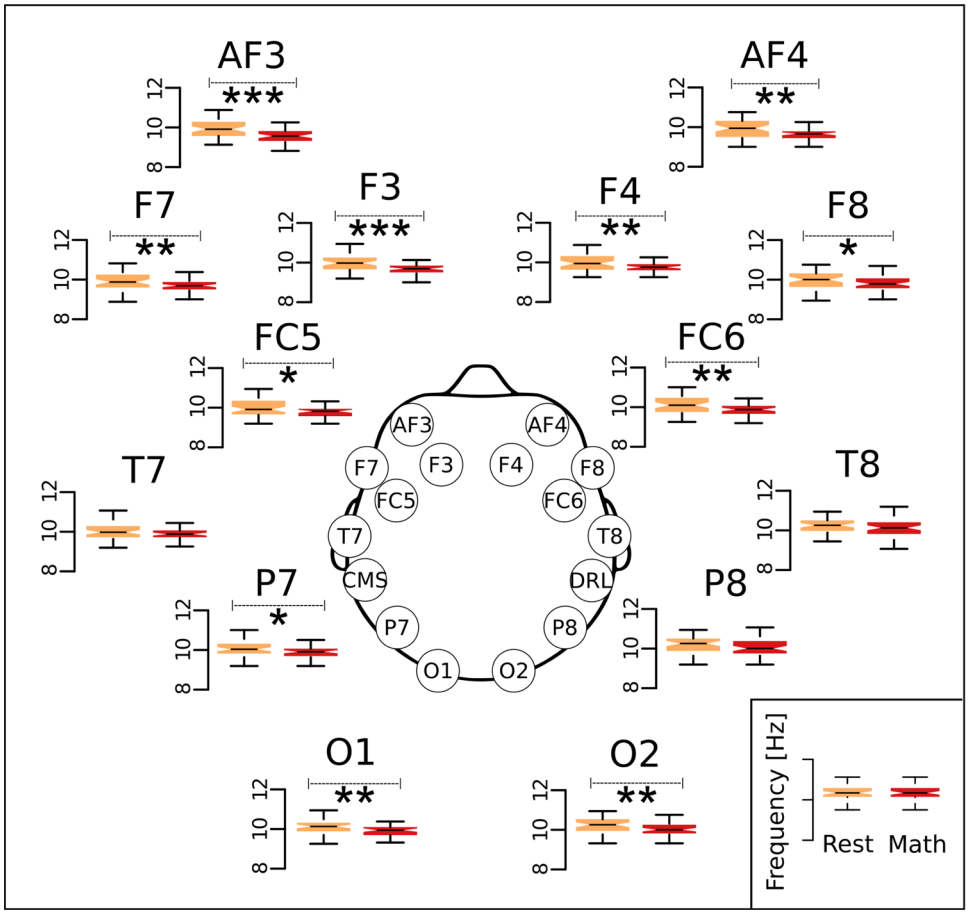

Fig. 3 MP atom central frequency (Freq in Hz) in the alpha range for Rest vs. Math. Two-tailed Wilcoxon rank-sum tests. FDR-corrected p-values (q-values) $* \, p \leq .05$; $** \, p \leq .01$; $*** \, p \leq .001$

non-significant trend toward slightly longer atom length was present both on alpha and theta ranges during Math. There were no clear antero-posterior or inter-hemisphere distribution trends for theta or alpha length during Math.

3.6 Comparisons between conditions along the [0.5 Hz–20 Hz] range

When atom number and amplitude were examined along the [0.5 Hz–20 Hz] range in terms of the strength of the statistical difference between conditions, some interesting patterns concerning atom number comparisons were revealed. These results are shown in Fig. 4. It should be remembered that alpha count was lower and theta count was higher during the Math task. In the first place, theta range effects were dissociated from those on the alpha range and occurred in a continuum with the delta range. Second, theta count differences between conditions, where present in left frontal region, were significant along a diffuse range encompassing 3 Hz to 7 Hz frequencies. Conversely, alpha range differences were consistent on a narrow band around 10 Hz to 10.5 Hz. Third, theta range effects were notable on the left frontal region, whereas alpha range effects were more diffuse and most evident on the right hemisphere and on posterior sites, reaching significance on left frontal, right posterior temporoparietal, and bilateral occipital regions. Of note, the left frontal region, where atom count on the complete alpha (7.5 Hz–13 Hz] range was not significantly different between conditions, still contained a small [10 Hz–10.5 Hz] range with significant differences. Microbin power and MP amplitude comparisons between Math and Rest were non-significant.

4 Discussion

This study made use of a matching pursuit procedure to investigate EEG alpha and theta time-frequency structure during a resting baseline condition and a sustained written arithmetic task obtained with a streamlined experimental protocol, outside the EEG lab, with the aid of a mobile EEG device.

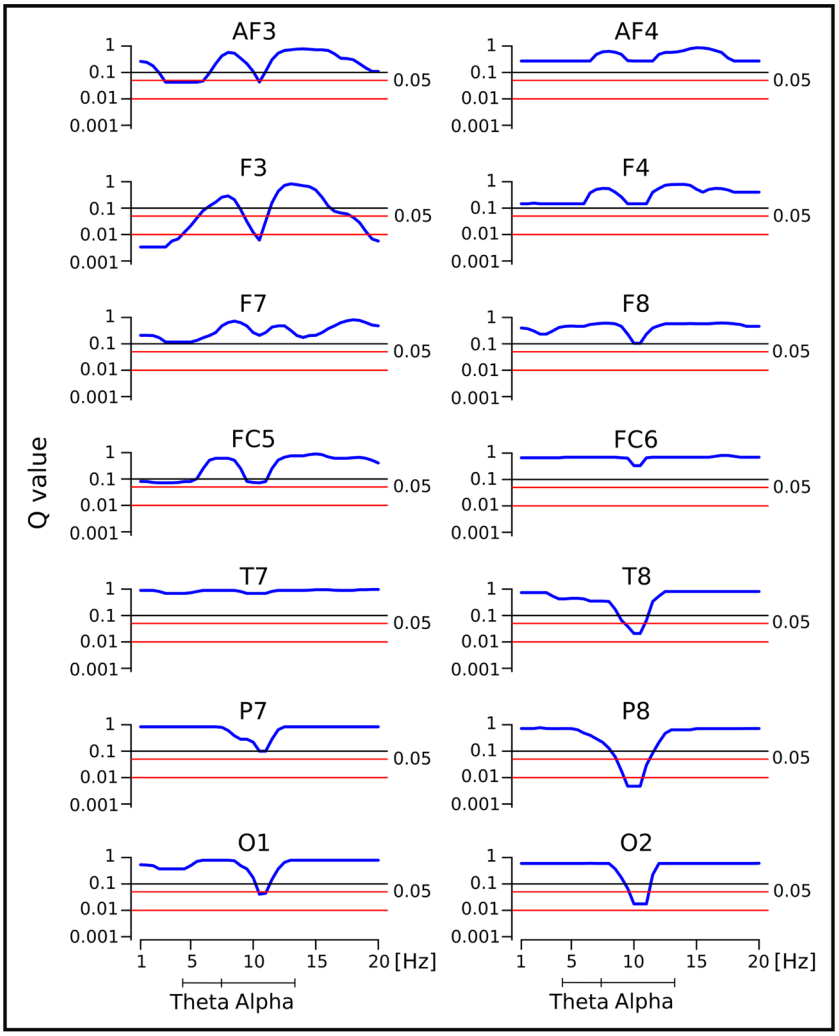

Fig. 4 FDR-corrected p-values (q-values) for two-tailed Wilcoxon rank-sum test MP atom number comparisons between Rest and Math conditions along the [0.5 Hz–20 Hz] range, in 0.5 Hz bins, according to electrode distribution. Y axis in logarithmic scale. Significance thresholds 0.05 and 0.01 respectively depicted in red and black. MP: Matching Pursuit; MP atom number: atoms/min. See text for details

Relative to resting wakefulness, there was lower MP alpha count over posterior and right-side regions during the Math task, whereas MP theta count was significantly higher on anteriorly located electrodes, especially over the left hemisphere. We interpreted these results as indicating alpha and theta activation during the Math task, in line with previous findings obtained with other signal analysis techniques and considering different time frames [66–68].

4.1 Alpha and theta activation during Math task

In [66], 0.78 Hz bands were analyzed in 1280 ms EEG segments with frequency domain variable resolution electromagnetic tomography. During an arithmetic production task for 10 subjects, higher primary field currents were found for the 3.9 Hz mini theta band over left inferior frontal gyrus and 5.46 Hz mini theta band over right prefrontal cortex, whereas the 12.48 Hz mini alpha band showed a decrease over left parietal cortex. In [67], 8 s EEG epochs obtained from 18 medical students during two-digit multiplication relative to viewing numbers showed increased absolute theta power in the frontal region and decreased absolute alpha power in the parietal region. In [68], during each first second of post stimulus activity, divided into 250 ms epochs, upper alpha ERD was stronger over parietal vs. prefrontal, and theta ERS stronger on prefrontal vs. parietal regions during implicit retrieval of multiplication fact knowledge, tested on 17 university students. Our findings of most significant left-sided theta and right-sided alpha effects are also consistent with long-range anterior-posterior and inter-hemispheric functional connections between several different neuronal populations [8, 9]. Theta and alpha activity obtained during cognitive and motor tasks result from different as opposed to unitary sources on frontal and central cortex [69–71]. Our results are also congruent with functional neuroimaging studies that repeatedly implicate a fronto-parietal network subserved by bilateral intraparietal and prefrontal cortex on Math reasoning and cognition [8, 9, 72].

Alpha count differences between conditions were particularly noticeable on a narrow range around 10 Hz–10.5 Hz, consistent with narrow alpha peak differences shown in ERD/ERS studies [73]. On this range, differences were also present on left frontal sites, where entire alpha range comparisons were non-significant. This reinforces the importance of investigating peak alpha frequency and narrow, as opposed to traditional alpha bands. We also found widespread lower median alpha frequency during the Math task relative to resting wakefulness, as the proportion of lower frequency alpha atoms increased in association with a total decrease on alpha atom number, consistent with the concept of alpha activation.

Alpha and theta activation are considered to reflect different cognitive functions related to Math problem solving and overall cognitive processing [10, 12]. In the above-cited study by De Smedt and cols, higher left hemisphere theta ERS was associated with solution of small-sized arithmetic problems and judged to reflect fact retrieval functions, whereas stronger bilateral alpha ERD was associated with solution of large-sized problems and considered to reflect more intense engagement of cognitive resources [12]. Conversely, Klimesch and cols proposed that "theta activity reflects working memory functions whereas upper alpha activity would be related to reactivation of long-term memory codes during short-term memory processes" [73]. Because our Math task required different cognitive abilities assumed to overlap and rapidly alternate over time, no theoretical assumptions regarding cognitive strategy or specific cognitive functions may be attempted [74]. Indeed, due to the lack of a control non-Math cognitive task, it is not possible to specifically attribute our results to mathematical reasoning, as they might be non-specifically related to cognitive engagement in comparison to a resting period. Increased frontal theta and decreased parietal alpha activity, as measured through relative band power, have been verified in a study that tested different mobile EEG devices comparing a more demanding with a less demanding cognitive task [23]. The four-by-four-digit multiplication task employed in the present study required the completion of sequential addition steps and most likely recruited attentional resources as well as rote memory functions, relying on variable verbal, visuo-spatial and quantitative processes across subjects [51, 75]. Functional neuronal networks are believed to exhibit dynamic interactions between oscillatory activity produced in different frequency ranges, in other words, neuronal activity synchronization and desynchronization patterns in different frequency bands is believed to be related to the coupling and uncoupling of dedicated functional networks [76–80]. It has been shown that the sense of quantity is used abstractly by different neural networks in the brain [81–83]. Brodmann area 39, topographically related to the angular girus, has been considered as a boundary between low theta and high alpha frequency regions in EEG acquisition [34, 75, 83] and mental calculations have been claimed to be the most efficient way to activate Brodmann area 39 and adjacency [4, 8, 82, 83]. It is also known that oscillatory activity desynchronization on the alpha range reflects different functional processes over different scalp locations, for instance, suppression of mu rhythm over rolandic cortex is related to motor action and/or intention [84]. It should be noticed that so-called alpha activation as shown through ERD, spectral power attenuation or MP atom number reduction might indirectly reflect a contribution

of higher frequency, gamma activity synchronization over locally dedicated workspaces rapidly changing over time [78]. In any case, due to the plasticity of the network, some features of a dedicated workspace may be self-supporting and remain active for a relatively longer time frame, even after task completion [79]. Future studies might show whether dynamic inter-relations of MP atoms in alpha, theta and other frequency bands might be informative in this way. Perhaps the most distinctive aspect of mathematical language is that, unlike verbal language, its use absolutely requires precision. It may be intriguing that this need for accuracy is met by such a widespread instead of highly localized functional network, but that is precisely the concept conveyed in models addressing effortful cognitive tasks [72, 79, 85].

4.2 Matching pursuit analysis, signal non-stationarity and inhomogeneity

MP atom number translates decomposition of a strongly non-stationary signal into numerous elements that, in this case, fitted the time frame of several seconds. The fact that median atom duration was relatively stable across scalp topography and showed minor changes during the Math task may be relevant from the physiological standpoint, as the time frame implicated in ERD/ERS effects lies within the range of a few seconds [36]. However, estimations of MP atom duration depend on the employed methodology [86]. This study made use of a low-sized stochastic MP dictionary. Increasing size of the stochastic dictionary will consistently reduce atom duration, as reported by Chandran and cols [86]. Of notice on that study, the ultimate dictionary size was chosen to fit what was a priori considered an optimal gamma burst duration, as measured through other techniques. Our choice of dictionary size finds support on one high spatial-resolution EEG study of ERS/ERD involving 20 healthy university students, where individually tailored activation periods were found to range approximately between 1 s and 3 s [12]. Another finding that may be directly related to signal inhomogeneity was an unexpected dissociation between amplitude and MP atom count. No significant changes were found for MP atom and power amplitude during our Math task, despite consistent inter-hemisphere differences in these quantities. Arguably, our spectral analysis spanned long signal sections, increasing variance and potentially reducing statistical differences. However, similar results for MP atom amplitude render this explanation less plausible. Atom number may be more sensitive than amplitude to detect the functional interplay of alpha and theta oscillations during a cognitive task. On a previous investigation of resting alpha activity in a different study sample, we have reached a similar conclusion about the advantage of atom number computation over amplitude or duration measures to discriminate between eyes open and eyes closed states [49]. In the present study, amplitude-related findings are consistent with reports that, during a sustained cognitive task, ongoing (tonic) EEG activity resembles baseline rather than phasic (event-related) activity [20, 39]. Additionally, our divergent findings on the topographic distribution of atom number and amplitude-related quantities further suggest a functional dissociation between these variables.

The main objective of implementing MP signal analysis for the study of Math cognition in a realistic setting was successfully carried out on an off-line basis. Time and computational cost for an online procedure remain to be verified [22].

4.3 Study protocol for data acquisition

Another study aim was to test the feasibility of a simplified experimental protocol for data acquisition. Subjects were wearing a mobile EEG device. They were tested outside the EEG lab, in the sitting position with eyes open. They performed the written Math task during a relatively prolonged period of time, consisting of a whole minute to carry out each calculation. Our results suggest this experimental setup was overall successful. However, important limitations for data analysis and interpretation ensued from the experimental setup. Blinking and eye movement artifacts were not controlled between conditions, so data on lower (delta) frequency bands was not analyzed. The written task may have confounded results over EEG central and parietal areas, as hand movement and motor intention are known to influence alpha range activity (mu rhythm), particularly picked up over C3 and C4 electrodes [35, 87], and alpha band activity may be suppressed on both hemispheres during writing, mainly over the left parietal lobe [88]. This protocol may be more suitable to explore anterior frontal and temporo-occipital regions, whereas depending on the study aim, central and parietal regions might be more appropriately surveyed with non-motor tasks. Future studies should more carefully control for movement and eye blinking, and differentiate between Math tasks and other cognitive functions.

5 Conclusions

In conclusion, with the aid of mobile EEG, it was possible to test feasibility of a simplified experimental protocol to investigate EEG alpha and theta activation during sustained written arithmetic calculations outside the EEG lab, employing for the first time a matching pursuit procedure for

EEG signal analysis in this context and replicating previous results obtained with different techniques and different time frames. Over the relatively relaxed time frame of 60 s, MP with a low density-sized stochastic dictionary loosely decomposed alpha and theta activity into numerous elements with duration around 3 s, similar to the duration of tightly time-locked activity obtained with ERS/ERD studies. MP alpha and theta atom number and central alpha frequency were informative to document regional activation during the Math task. MP atom number appears to be more informative than amplitude in this type of experimental paradigm. Future studies employing higher resolution equipment, more specific cognitive tasks and better exploring spatial and temporal dynamics of MP alpha and theta activity may prove useful, but the goal of achieving valid experimental results on realistic settings remains a priority. MP merits to be further explored in the context of Math cognition studies.

Appendix A

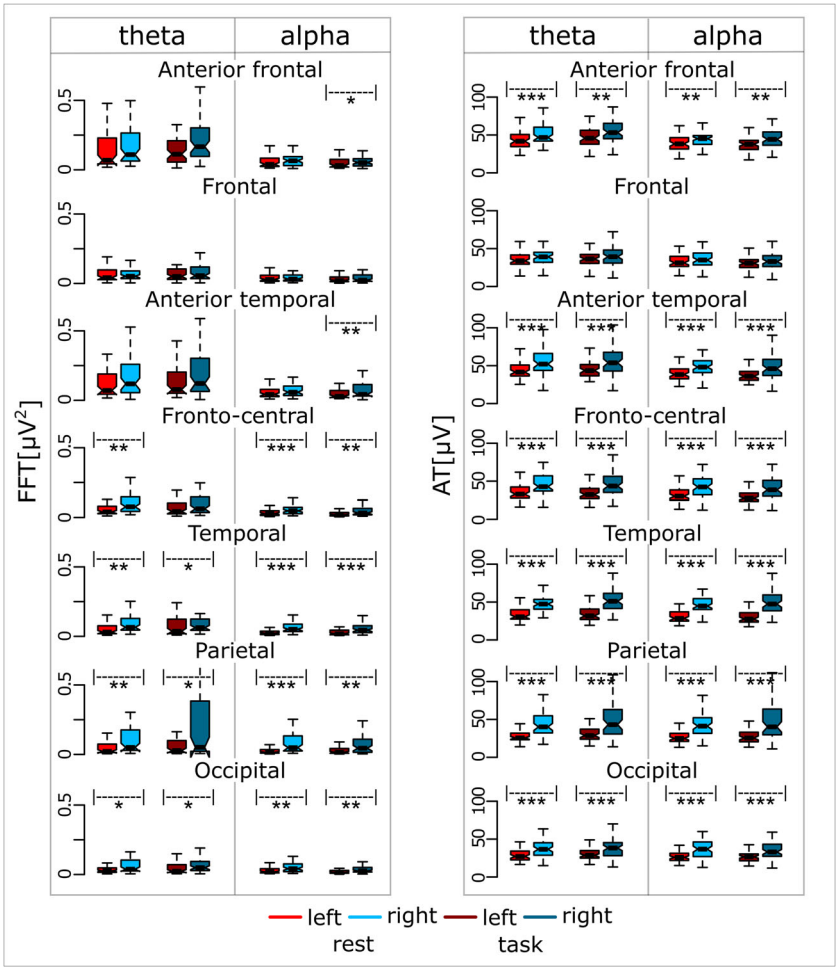

Fig. 5 Topography distribution and inter-hemisphere comparisons for amplitude-related quantities, spectral power and MP AT, during Rest and Math (called task in Figure). Notice the similarity of behavior between conditions and the greater significance of AT compared to Power. Two-tailed Wilcoxon rank-sum tests. FDR-corrected p-values (q-values) $* \leq .05$; $** \leq .01$; $*** \leq .001$. AT: Atom amplitude; MP: Matching Pursuit; Power: average absolute band-limited spectral power

Appendix B

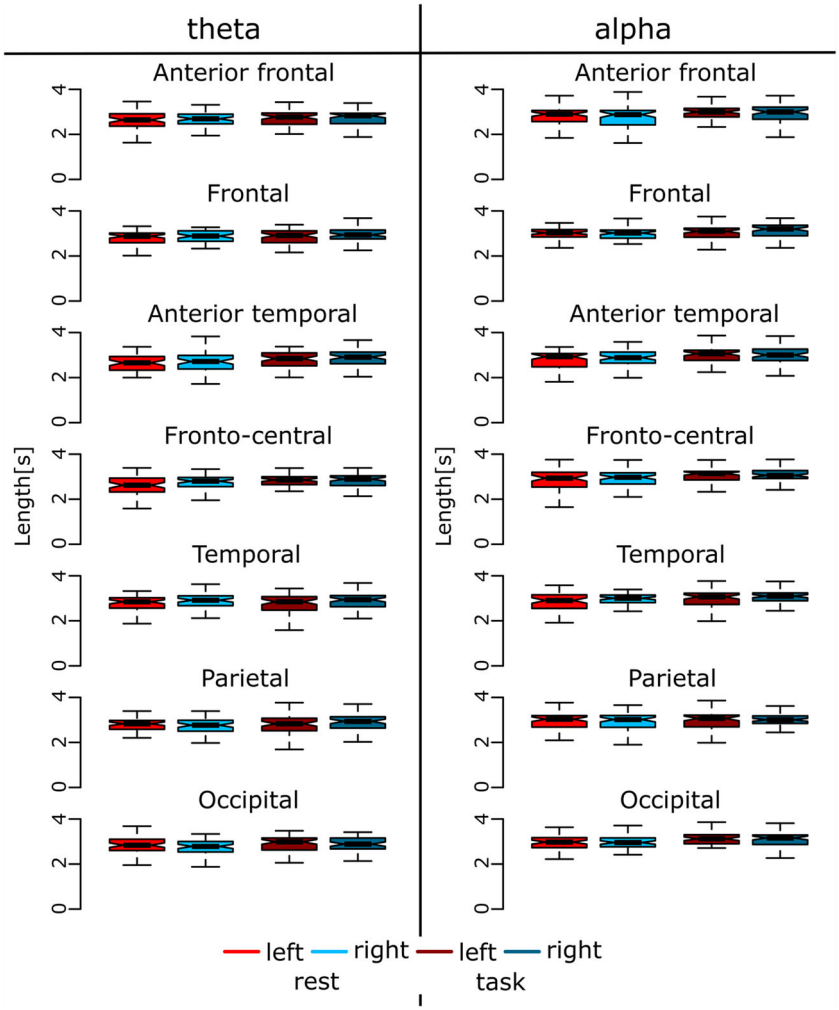

Fig. 6 Topography distribution and comparisons between Rest and Math (called task in Figure) conditions for MP atom duration. Inter-hemisphere comparisons were non-significant. Two-tailed Wilcoxon rank-sum tests. FDR-corrected p-values (q-values) $* \leq .05$; $** \leq .01$; $*** \leq .001$. MP: Matching Pursuit. See text for details

Author Contributions GJLG: data curation, software, methodology, formal analysis, writing — original draft; GMB: data curation, software, methodology, formal analysis, writing — original draft; GMJ: data curation, investigation; ALM: data curation, investigation, statistical analysis; JLRF: conceptualization, funding acquisition, investigation, methodology, project administration; LMAO: data curation, investigation; SVS: data curation, formal analysis, methodology, writing — review and editing.

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Declarations

Conflict of interest The authors declare no competing interests.

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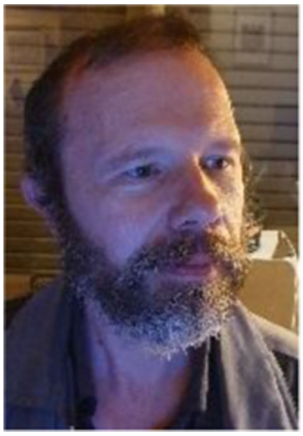

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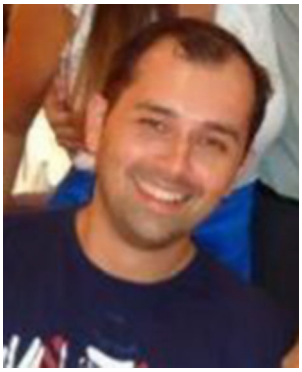

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